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Enhancing lung diseases recognition through CNN-RNN methodologies

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Diagnostics of respiratory disorders greatly benefit from medical imaging, especially X-ray imaging, which offers important information about the anatomical anomalies of the lungs. As we delve deeper into the field of lung illness recognition, it becomes clear that utilizing multiscale Deep Convolutional Neural Network (DCNN) techniques has the potential to transform the detection of pneumonia and tuberculosis from X-ray images. In this paper, we will classify images through a process that requires only chest X-ray images. We have proposed a deep learning (DL)-based algorithm for lung disease detection, which we term the Convolutional Recurrent Network (C-RNet). In our research, we classify CXR images into four categories according to the publicly available dataset. Our proposed model can calculate the dependency and continuity properties of the intermediate layer output very precisely. At the same time, the features of these intermediate layers can be combined with the final fully-connected network for classification prediction, resulting in better classification accuracy. We have explored the potential of combining CNN and RNN with XAI to identify lung diseases from chest radiographs to improve diagnostic accuracy compared to traditional single-scale methods. Upon comparing our suggested model with the current models, we discovered that, with an accuracy of 93.73%, F1-score of 94.6%, total floating point operation per second (FLOPS) count of 637,222,592, total parameter count of 1,901,764 and model size of 7.25MB on the full dataset, our suggested model achieved the best accuracy of all the architectures we compared. Moreover, our suggested model, C-RNet, was observed to accurately categorize and detect the regions of disease through approaches such as Grad-CAM.

Keywords Lung diseases, X-ray, CNN, RNN, XAI, GRAD-CAM, LSTM.

Motivation

Lung diseases rank among the most infectious health conditions and have become the leading causes of mortality worldwide. Developing countries like Bangladesh are particularly vulnerable to these diseases due to environmental hazards. (Fig. 1) These diseases stem from various causes. Exposure to pollutants, dust, chemicals, infections, and smoking can cause lung diseases like pneumonia, asthma, tuberculosis, influenza, COVID and other chronic obstructive pulmonary disease. Due to the rising of air-pollution, the number of respiratory diseases is also increasing. Prolonged exposure to polluted air often leads to chronic respiratory conditions, weakening individuals over time. The overall number of patients suffering from respiratory and infectious diseases in Bangladesh has increased drastically over the previous five years[1]. Experts warned that the infection could be disastrous for individuals older than 50, since over 70% of Bangladesh's elderly population already has comorbid conditions and non-communicable diseases[1].

Pneumonia killed over 740,180 children in 2019 and is responsible for 14% of mortality among children below five. Annually, pneumonia claims the lives of about 12,000 children in Bangladesh under five[2]. In Bangladesh, pneumonia causes the deaths of about 24,000 children annually, making about 24% of all deaths among children under five, which is higher than the global scale average[3]. This means that two to three kids pass away from pneumonia per hour. Even with these concerning figures, only 60% of parents take their child to the doctor when they show signs of pneumonia. 40% of children do not obtain necessary treatment since there is a lack of awareness[3]. Moreover, lung diseases can interrupt the vital process of people's respiratory systems,

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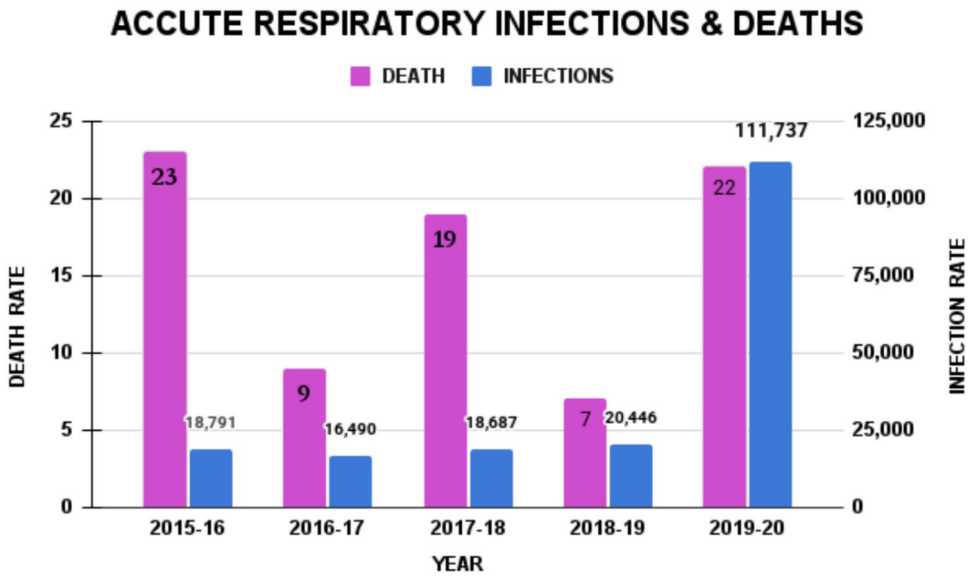


Fig. 1. Increasing rate of resperitory diseases[1].

causing uncomfortable symptoms, such as fatigue, wheezing, coughing, shortness of breath and chest pain. The signs and symptoms of these diseases are quite similar. Hence, it is challenging to diagnose different diseases and critical to identify and treat them early to enhance patient outcomes and lessen the burden of these conditions on society. Addressing this growing health crisis requires urgent action, from reducing pollution levels to enhancing diagnostic techniques and treatment accessibility.

X-ray radiography is frequently the favored method for identifying lung diseases, as it is both feasible and inexpensive. While CT scans provide higher detection sensitivity, X-ray radiography remains the preferred choice in clinical settings due to its affordability and widespread accessibility in general hospitals[4]. Traditional lung disease diagnosis relies mainly on a radiologist's ability to analyze radiological images. Examining and analyzing medical images requires a reliable diagnosis since symptoms and imaging characteristics can overlap among different lung diseases. In order to prevent important information from being overlooked, precise and careful attention is required. Furthermore, the lack of qualified radiologists in many areas is increasing the need for diagnostic services. This further emphasizes the improvement in testing and medication to assist radiologists and accelerate the identification process. Researchers have built deep learning algorithms that make use of advanced machine learning techniques and neural networks. These models or algorithms assess and classify the X-ray pictures according to unique patterns and characteristics linked to different lung diseases.

Our suggested model also utilized advanced machine learning algorithms. We have developed our model using a number of techniques based on the application of CNN and RNN methodologies to CXR data. To train our C-RNet model, we have used *COVID-19 Radiography Database* dataset[5,6]. The dataset contains CXR images of COVID-19, Lung Opacity, Normal and Viral Pneumonia cases. We integrated RNN methods like Bi-LSTM techniques with CNN and used XAI methods like Grad-CAM (Gradient-weighted Class Activation Mapping) to develop our model. Subsequently, we have recorded every step of the procedure, including gathering data, preprocessing, training the model, evaluation and interpreting the outcomes. Our proposed model, C-RNet, aims to bridge the gap between cutting-edge AI methodologies and real-world clinical requirements by integrating multiscale CNN, RNN and XAI components.

Problem statement

Existing deep learning models for lung illness diagnosis still have serious drawbacks, despite notable developments in medical imaging and AI-driven disease classification. When it comes to extracting spatial data from chest X-ray (CXR) pictures, CNNs have shown remarkable performance. Nevertheless, they frequently overlook contextual information and sequential dependencies that could improve diagnostic precision. When it comes to identifying minor illness progressions, when temporal or region-specific correlations are critical, this constraint is especially troublesome[7].

Additional issues, including dataset imbalance, noise in X-ray pictures and lack of interpretability further impede AI-based diagnosis in practical applications. Biased model predictions result from the excessive number of normal cases in large medical datasets relative to ill samples[8]. Furthermore, model generalization is challenging due to X-ray pictures' frequent artifacts, illumination fluctuations and inconsistent scanning techniques across various healthcare facilities. A significant obstacle to clinical adoption of deep learning models is their explainability, since medical practitioners find them less reliable when they provide black-box predictions without explanation.

In order to overcome these difficulties, this study integrates sequential pattern recognition and spatial feature extraction to develop a hybrid CNN-RNN framework (C-RNet) for lung disease classification. By showing the model's decision-making process, the use of XAI approaches like Grad-CAM improves clinical interpretability

and transparency. Our method seeks to provide greater accuracy, improved generalization and increased reliability in practical medical applications by optimizing model architecture based on assessment metrics and training efficiency.

Research contribution

This research provides a valuable solution through multi-scale CNN and RNN methodologies to identify lung diseases from X-ray images in developing countries, including Bangladesh, where radiologists are scarce in the medical field, especially in rural or remote areas. Multi-scale CNN is utilized for effective feature extraction at different dimensions to enhance the identification and categorization of lung disorders, while Bi-LSTM networks are employed to identify spatial patterns and related dependencies for more precise disease identification. Moreover, the model's ability to utilize multi-scale processing enhances its capacity to distinguish and classify among multiple lung diseases, resulting in increased diagnosis accuracy when compared to conventional single-scale approaches. Furthermore, including XAI methods in the suggested model will enhance its credibility and comprehensibility. Grad-CAM was used to visualize and understand the process of how the model makes decisions.

The primary objective of this research is to develop an enhanced deep learning model that improves the accuracy of lung disease detection using CXR images. To achieve this, we set the following specific objectives:

- We aim to develop an advanced patch-based augmentation technique that improves generalization across different imaging conditions without any metadata.
- Our model is designed with computational efficiency in mind, making it more suitable for real-world deployment in clinical settings.
- Implementing an optimal architecture selection strategy for C-RNet, balancing accuracy, training efficiency, and interpretability.
- Integrating explainable AI (XAI) techniques like Grad-CAM to improve model transparency, aiding in clinical decision-making and trustworthiness.

The C-RNet approach aims to assist clinicians in diagnosing patients more quickly and correctly by presenting them with clearly understood data. It assures that the results are clear, making it easier for radiologists and other healthcare professionals to understand the diagnostic insights, hence increasing dependability and confidence. This advancement will increase the overall reliability of medical processes. The fundamental goal of this research is to improve the efficacy and efficiency of correctly detecting lung diseases.

Paper organization

The remaining sections of this work are organized as follows. Literature Review provides an in-depth overview of relevant work, concentrating on existing CNN and RNN architectures, Grad-CAM approaches, and past fusion models, while describing the reason for our research. Methodology discusses the workflow, data description, preprocessing procedures, and the description of our C-RNet model. The experimental evaluation describes experimental setup, evaluation metrics, performance analysis, impact of dataset imbalance and results comparison, as well as Grad-CAM-based visualization. Finally, the discussion part addresses key findings, highlighting the favorable aspects of our proposed model above others, contextualizing the findings and offering recommendations for further research, and is followed by the reference section, which lists all references utilized in this study.

Literature review

The goal of this review of the literature is to examine the developments in multi-scale deep learning approaches for detecting lung illness from CXR pictures. We selected a few studies, such as CNNs, RNNs, hybrid models, and XAI models, in order to analyze the accuracy, efficiency, and readability of different models. We have attempted to figure out the main obstacles to lung disease recognition using deep learning and investigate potential solutions. Furthermore, we have chosen research papers published between 2020 and 2024 to provide a thorough and current study that takes into account the most recent developments and patterns in machine learning for image analysis in medical imaging. We have investigated papers that present new architectures, optimal methods, or upgrade procedures, including multi-scale pattern extraction, data augmentation, and transfer learning.

Early CNN approaches for lung disease detection

Early research was mostly done on analyzing medical images using standard CNN architectures to classify chest-related diseases, especially COVID-19. Early research efforts in medical image analysis largely employed conventional CNN architectures for classifying chest-related diseases, particularly COVID-19. Bukhari et al.[9] utilized a ResNet-50-based CNN to differentiate between normal, pneumonia, and COVID-19 from a small dataset of 278 CXR images. While achieving satisfactory accuracy, the model's utility was constrained due to the limited dataset size and lack of validation on more diverse or larger datasets. Furthermore, it lacked the integration of interpretability mechanisms, which is critical for clinical adoption. Likewise, Atitallah et al.[10] developed the RND-CNN model to classify CXR pictures into COVID-19, pneumonia, and normal categories. This model was employed using four convolutional blocks and two fully connected layers. Despite promising results, the lack of testing across various imaging modalities and its incapacity to treat a broader range of pulmonary disorders limited its practical impact, despite encouraging results.

Researchers started adding deeper layers and stronger transfer learning techniques to standard architectures in an effort to further improve CNN performance. Patil et al.[11] created a modified VGG-19+ model for lung cancer detection using CT images and achieved 95% accuracy, demonstrating enhanced performance through

data augmentation and a trustworthy dataset (LIDC/IDRI). However, this model's real-world viability was diminished by its high computational requirements and lack of model explainability features. Gour et al.[12] suggested an alternative improvement by introducing UA-ConvNet. It was a CNN model that incorporated EfficientNet-B3 with Monte Carlo dropout to measure uncertainty in COVID-19 predictions. Nevertheless, its applicability was once more restricted to CXR images and lacked extension to CT data or wider disease coverage. Muralidharan et al.[13] made progress in multiscale feature learning by using their deep CNN model, which divided images into seven semantic modes without using peak identification in the Fourier domain. However, its performance was significantly reliant on manually selecting optimal mode combinations and the limited, imbalanced dataset. A limited examination using hold-out validation raises questions regarding the generalizability and automation of feature selection.

Some researchers have also explored various CNN-based and ensemble models to expand the range of diseases they can classify, all while enhancing the accuracy of their classifications. In order to classify CXR images into COVID-19, pneumothorax, pneumonia, lung cancer and tuberculosis, Malik et al.[14] introduced CDC_Net, an end-to-end CNN architecture that combines residual learning and dilated convolution. Strong generalizability and adaptability were shown by their ten-fold cross-validation, normalization and robust evaluation across twelve datasets.

Similarly, Ifty et al.[15] utilized a mix of CNNs (like Xception and DenseNet), ViTs and Big Transfer (BiT) for multi-class classification. They attained 96.21% accuracy using stratified 5-fold validation. They used XAI tools like Grad-CAM and LIME to shed light on how the model made its decisions, emphasizing just how crucial transparency and trust are becoming in the medical field. However, these CNN-focused models struggled to capture the contextual relationships across different areas in imaging data because they lacked spatial-sequential modeling. In short, without that spatial-sequential approach, these models fell short in understanding the context within the imaging data.

As deep learning progressed, ViT emerged as a powerful alternative to CNNs for medical imaging tasks. Dayan et al.[16] took a closer look at both full-image and lung-segmented ViT methods for classifying CXRs. Their findings were impressive, showing accuracy levels of up to 97.83% in three-class settings and high AUC scores in multi-class tasks. What's particularly interesting was that full-image ViTs can streamline preprocessing by eliminating the need for complex lung segmentation. However, when they tackled more challenging eight-label tasks, performance took a hit, raising concerns about how well these models can generalize as class complexity increases.

Islam et al.[17] suggested a hybrid method that combines deep CNNs with GRUs for the detection of both COVID-19 and lung cancer in order to close the gap in spatial and temporal modeling. Because of GRU's lightweight architecture and the incorporation of interpretability tools like LIME and SHAP, their approach shortened training time. When working with sequential or time-sensitive data, temporal explainability was still difficult to achieve. Our study aims to improve lung disease recognition using a CNN+RNN hybrid framework. These limitations have prompted more research into hybrid CNN+RNN models, which can effectively capture both spatial and sequential patterns.

Hybrid CNN–RNN architectures for sequential feature modeling

To address CNN shortcomings in handling sequential dependencies and global visual context, researchers suggested hybrid architectures that combine CNNs and recurrent neural networks (RNNs). Several recent studies have investigated the efficacy of hybrid CNN-RNN models for lung disease diagnosis, focusing on the strengths of CNNs for spatial feature extraction and RNNs for temporal or sequential learning. Khadhim et al.[18] suggested a VGG19-RNN model with 97.8% accuracy in categorizing COVID-19, pneumonia, and normal CXRs. This model stood out for its interpretability with Grad-CAM and a balanced dataset, but its small COVID-19 sample size raised questions about generalizability. Similarly, Kanjanasurat et al.[19] attempted to replace fully connected layers in ResNet152V2, VGG19, and DenseNet121 with LSTM and GRU layers. The CNN-RNN hybrid obtained 93.37 percent accuracy and prioritized preprocessing (CLAHE, gamma correction) to improve image quality. However, the solution required a lot of GPU and RAM, which limited its use on resource-constrained systems.

Jain et al.[20] improved the field by employing a multiscale CNN-RNN architecture trained on a larger and more diversified dataset, which included tuberculosis, lung cancer, and several kinds of pneumonia. Their model merged max-pooling CNN layers with LSTM units and used Grad-CAM to improve interpretability at many levels. While previous studies addressed the issue of limited work scope and lack of transparency, the complexity resulted in higher computational demands. Similarly, Mohammad et al.[21] presented CNN-GRU and CNN-LSTM frameworks, attaining up to 98.5 percent accuracy. Their model excelled at generalization through robust augmentation and added temporal processing to static images, but it was based on a single dataset, which increased the risk of overfitting.

Nair et al.[22] also proposed a CNN-LSTM study for the multi-class detection of lung diseases. They accurately diagnosed six lung conditions, achieving an impressive F1-score of 94% and precision of 95%. Their method incorporated SMOTE for class balancing and implemented a robust preprocessing pipeline. This study outperformed models such as DenseNet, InceptionV3, and gained trust as a clinical diagnostic tool. Despite its potential, this approach had many drawbacks, most notably its dependence on certain datasets and the tendency for deep sequential models to overfit. However, future studies in this area suggest emphasizing the development of specialized architectures and incorporating more diverse datasets to improve generalization and exploration of advanced hybrid architectures and transfer learning techniques.

Developments in CNN architectures for explainability

Numerous studies have clarified how different deep learning architectures approach CXR images to identify lung conditions. For example, Islam et al.[23] examined the potential of deep convolutional networks on multiple

datasets. On the Shenzhen dataset, they discovered that employing ensemble DCNs improved accuracy, especially for detecting cardiomegaly, with a performance rise of 17%. Similarly, Sharma et al.[24] introduced two CNN versions for the detection of pneumonia and discovered that performing data augmentation and adding dropout layers greatly improved test accuracy from 74% to 90%. Their results indicated that strategies like batch normalization and preventing overfitting are crucial when working with smaller datasets. Other research had concentrated on creating or assessing AI models with varied datasets to enhance the detection of pneumonia and tuberculosis (TB). Abubakar et al.[25] pointed out that CNNs surpass traditional methods in TB classification, stressing the importance of dataset diversity by utilizing collections from Montgomery, Shenzhen, and Kaggle. Their comparative study between DenseNet-169 and SVM-based approaches highlighted how the size of the dataset greatly influences both credibility and accuracy. Kundu et al.[26] introduced a computer-aided diagnosis (CAD) system that merged GoogLeNet, ResNet-18, and DenseNet-121 through ensemble and transfer learning techniques. An example of a pneumonic lung X-ray alongside a healthy one from their dataset is depicted in Fig. 2. Their model achieved remarkable accuracy and sensitivity on the RSNA and Kermany datasets, showcasing the power of model ensembles.

The impact of deep learning on diagnosing COVID-19 and other respiratory illnesses has really started to take off. On the other hand, Bhosale et al.[28] used CNNs for feature extraction, followed by support vector machine (SVM) classification and logistic regression to detect COVID-19. Their approach cleverly combined cloud-based tools and image segmentation, achieving an impressive 95% classification precision. These results highlight just how crucial preprocessing, model design, and cloud deployment are for real-world applications.

Recently, researchers have started to dive into more advanced architectures and hybrid models to improve diagnostic robustness. Duong et al.[29] applied transfer learning by using pre-trained weights in PyTorch. Used several methods to augment the data for better robustness and achieved the best results with the modified Hybrid EfficientNet with ViT. Similarly, Ahmed et al.[30] noted that combining CNNs with deep learning architectures could assist in distinguishing between pneumonia and TB and reduce false positives. Their model demonstrated how machine learning may be realistically incorporated into clinical workflows by accounting for the quality of annotations, dataset fluctuations, and conventional diagnostic techniques.

Broader imaging and classification techniques have also been examined in other investigations. One of them was Ahmed et al.[31]. To improve classification accuracy, they experimented with transfer learning, local binary patterns, and intensity distribution remodeling. The practical difficulties of implementing AI in healthcare are reflected in their emphasis on handling massive image data. In their investigation of U-Net-based segmentation for TB detection, Nafisah et al.[32] introduced a system for TB detection by connecting deep CNNs, segmentation, and Grad-CAM to help in interpreting CXR images for diagnosis. Yet, their model only performed binary classification (TB versus non-TB) and did not use sequential methods or combine data features across various scales.

Lastly, two novel architectures for lung segmentation and tuberculosis classification, TB-UNet and TB-DenseNet, were introduced by Iqbal et al.[33]. Their CAD method integrated attention and dilated fusion blocks. Across three CXR datasets, this study demonstrated amazing segmentation quality and attained impressive accuracy, reaching as high as 97.7%. They greatly enhanced feature extraction and interpretability by combining DenseNet-169 with gradient-based visualization approaches. This study demonstrated how specialized architectures, supported by extensive datasets and careful model construction, could significantly improve medical image classification.

In conclusion, earlier studies have established a strong basis, demonstrating the effectiveness of deep learning in identifying lung diseases, presenting CNN–RNN hybrids for spatial–sequence modeling, and advancing XAI technologies. However, there is still a significant gap in developing a cohesive system that combines three essential components: precise explainability, sequential reasoning, and robust feature extraction. By using Grad-CAM to create interpretable maps, expanding hybrid CNN–RNN models to address a variety of lung conditions, and evaluating our system on a sizable and varied dataset, our study seeks to close this gap. By clarifying the reasons behind the model's predictions, this comprehensive approach not only improves classification performance but also fosters clinical adoption, increasing the transparency and trustworthiness of medical AI systems.

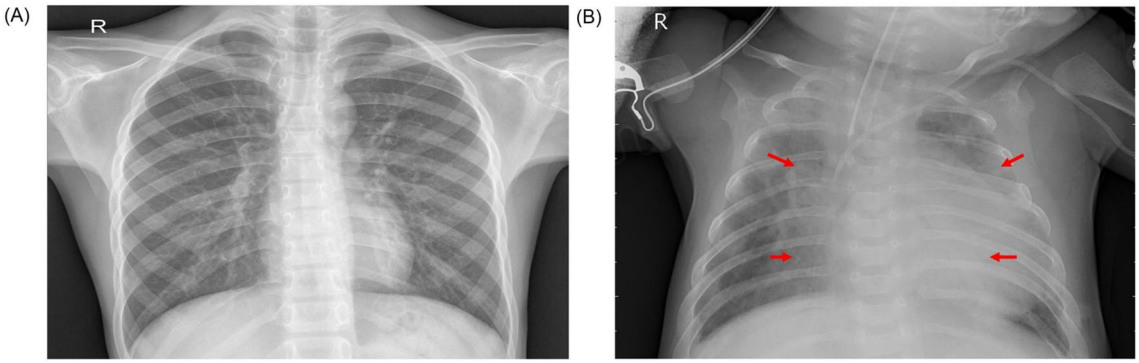


Fig. 2. Examples of two X-ray plates that display (**a**) a healthy lung and (**b**) a pneumonic lung[27].

Methodology

Workflow

In our paper, we aimed to develop a hybrid model combining a CNN-RNN architecture to enhance the overall performance for lung disease categorization based on CXR images. In Fig. 3, we walked through the steps of how we achieved our goal. In the initial step, we began by collecting our dataset. For data collection, we utilized the COVID-19 Radiography dataset and other publicly available CXRs, where each image was labeled and then categorized into classes, such as COVID-19, Lung Opacity, Normal, and Viral Pneumonia.

Next, in the data pre-processing phase, we resized our data to maintain consistency, normalized it to improve model convergence, and applied rescaling zooming to avoid overfitting issues. We also addressed missing values to avoid any bias in the model's learning process. During data pre-processing, we extracted patches from images to improve model performance. Additionally, we divided the images into small patches for the RNN branch. Finally, we split the data for training, testing, and validation, and by following these steps, our dataset was ready for further model training.

Following pre-processing, the images were passed into the proposed CNN and RNN model and finally into our proposed CNN + RNN hybrid model, C-RNet, which integrated spatial features from images and temporal features from sequences effectively so that our model could perform well on new, unseen, and robust data. The custom model then went to the training and evaluation phases, where the performance was assessed using accuracy, F1 score, recall, precision, training time, and other basic metrics. Once evaluated, the trained model was used for classification, which categorized input CXR images into one of the following classes: COVID-19, Lung Opacity, Normal, or Viral Pneumonia. We also used Grad-CAM to highlight the most important regions in the images that influenced the model's decision-making. Finally, our trained models were compared with Xception, Vision Transformer (ViT), VGG19, and Inception + LSTM architectures, which ultimately helped us identify the best-performing model, wrapping up the workflow.

Data description

We used *COVID-19 Radiography Database* dataset[5,6] containing CXR images of COVID-19 and other lung diseases. Developed in collaboration with Qatar University, the University of Dhaka, and institutions from Pakistan and Malaysia. This database aims to facilitate impactful scholarly work and AI-based advancements in COVID-19 detection and other pulmonary conditions. It is notable for being awarded the COVID-19 Dataset

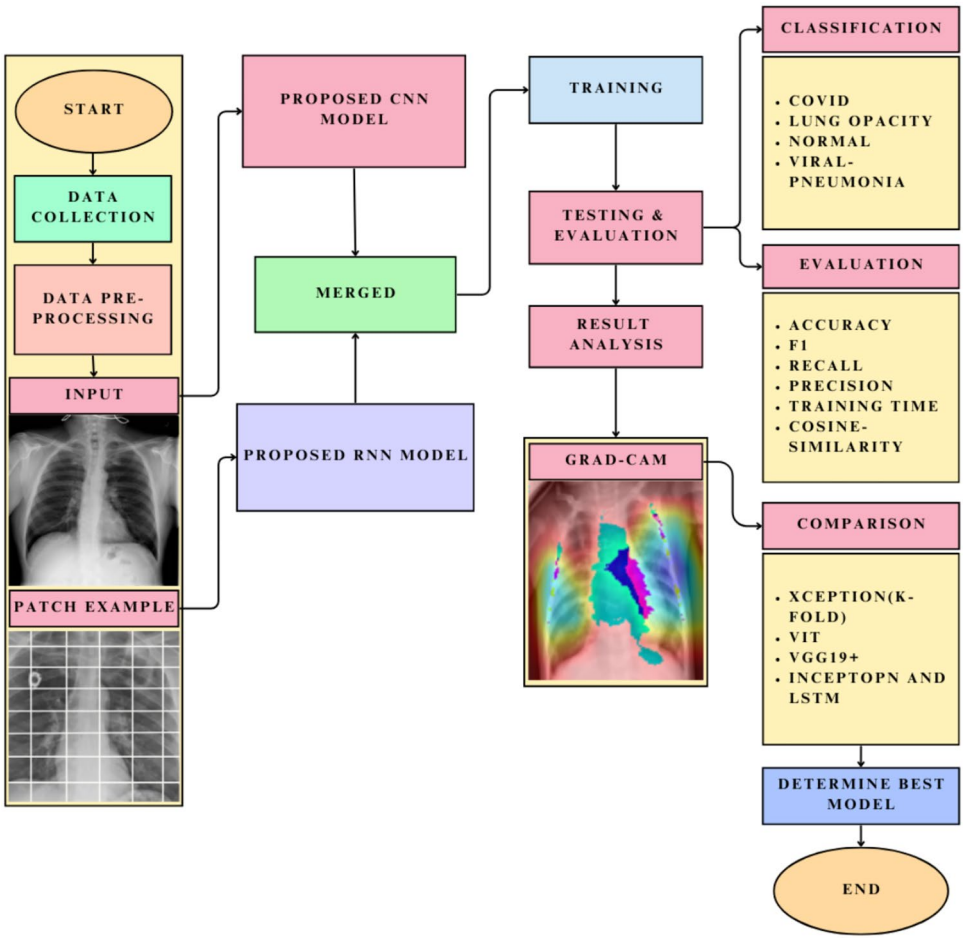


Fig. 3. Workflow of proposed C-RNet model.

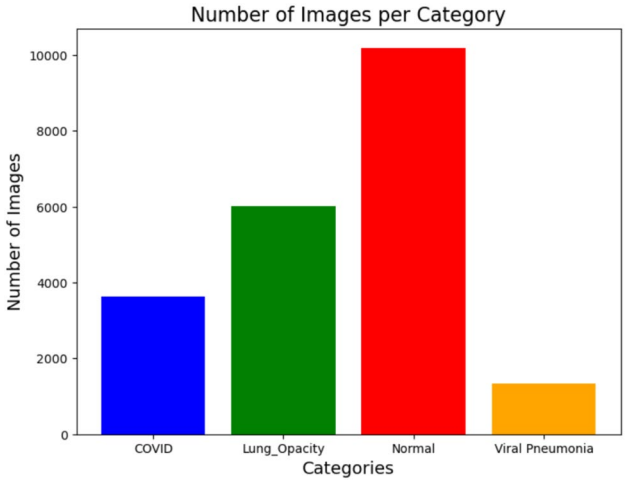
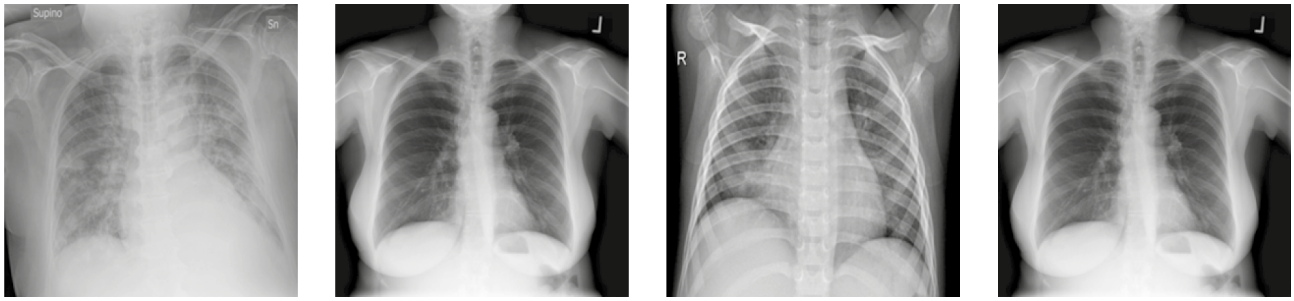


Fig. 4. Classification of dataset categories.



(a) Covid **(b)** Normal **(c)** Viral Pneumonia **(d)** Lung Opacity

Fig. 5. Visualization of dataset images.

Award by the Kaggle Community, highlighting its utility and importance in research. The dataset is categorized into four basic classes-COVID-19, Viral Pneumonia, Lung Opacity, and Normal.

The first version had 219 COVID-19 cases, 1,341 Normal cases, and 1,345 Viral Pneumonia cases. Currently, the dataset has 21,165 chest X-ray images. To offer an understanding of the dataset used in this research, Fig. 4 presents a classification chart summarizing all accessible data types, and Fig. 5 showcases sample images from the dataset, representing different lung conditions, including COVID-19, normal lungs, viral pneumonia, and lung opacity. The chart illustrates the distribution of data across various categories. This structural view is essential to ensure a clear representation of images during the model training phase.

Data pre-processing

We performed a number of pre-processing techniques, such as image resizing, labeling, scaling, dropout handling, and patch creation, to get the dataset ready for efficient model training. Initially, every image was resized to a fixed 128 x 128 pixel size. Images that returned None or did not resize were dropped. The following categorical labels were then applied: Viral Pneumonia = 3, Normal = 2, Lung Opacity = 1, and COVID = 0. After that, the dataset was divided into training and test sets, with 30% going to testing and 70% going to training. We rescaled image pixel values to fall between 0 and 1 in order to prepare the data for the convolutional layers. Each image was further separated into 8x8 patches because the model also incorporates an RNN component. By allowing the model to concentrate on specific areas of the image, these patches increase the accuracy and efficiency of learning. Furthermore, because patching naturally adds variation to the training data, it eliminates the need for data augmentation. The dataset was guaranteed to be well-structured and prepared for input into the deep learning model by this extensive pre-processing pipeline.

Model description

The C-RNet architecture is a custom-designed, dual stream deep learning model to detect lung diseases from cxr images. The motivation behind the proposed CNN–RNN hybrid architecture is to use parallel processing pathways to leverage the various ways that chest X-ray images can be represented. CNN is well-established at collecting hierarchical and localized spatial information. However, their receptive fields can occasionally limit how effectively they represent long-range spatial dependencies across distant regions of an image[34]. To overcome this limitation, a recurrent branch is utilized to operate on a structured sequential representation obtained from

the same input image. The recurrent branch for sequential dependency modeling and CNN for spatial feature extraction to process images concurrently.

CNN branch: spatial feature extraction

- *Input and Normalization* The CNN architecture workflow extracts with an input image of size $I \in \mathbb{R}^{128\times128\times3}$, representing a color image. The input is normalized using a rescaling layer that scales pixel intensities from the original [0, 255] range to [0, 1], ensuring numerical stability and faster convergence.
- *Convolutional Blocks (Core)* After rescaling, CNN processes the images through a series of five convolutional blocks with increasing filter depths [32, 64, 128, 256, 512] to extract more complex features and the feature maps' spatial dimensions were preserved by maintaining padding throughout. Each convolutional block applies the following operation:

$$F_l = ((\mathrm{Conv}_l(F_{l-1}))),$$

- *Regularization and Pooling* After completing the convolutional layers, batch normalization was applied to normalize the output which accelerated and stabilized the training process. This is followed by max pooling and dropout for spatial downsampling and regularization. L2 regularization was used to reduce overfitting in all convolutional layers. To improve the model's non-linearity, the ReLU (Rectified Linear Unit) activation function was implemented. The model was subsequently downsampled using a 2×2 pooling window, which decreased the spatial dimensions while retaining important spatial properties, increasing its efficiency. To further combat overfitting, a dropout layer that randomly deactivated 30% of the neurons during training was added to protect the model.

$$F_l = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_l(F_{l-1})\right)\right),$$

The final feature map F_5 is compressed using global average pooling, producing a 256-dimensional spatial feature representation suitable for global reasoning.

$$\mathrm{v_{CNN}} = \mathrm{GAP}(F_5),$$

The v_{CNN} represented the final output of the CNN branch, a condensed feature vector ready to be integrated into the downstream tasks such as classification. Figure 6 provides an overview of the CNN branch of our model.

RNN branch: sequential and contextual feature modeling
Initially in the RNN branch, image was decomposed into smaller non-overlapping patches to analyze the spatial features of the image. Each input sample was divided into 256 patches, each containing 192 features (specifically, the RGB pixel values of each patch).

$$I \to P = \{p_1, p_2, \ldots, p_{256}\}, \quad p_i \in \mathbb{R}^{192}$$

In this context, each patch p_i was treated as a timestep, enabling the RNN to learn inter-patch dependencies, effectively modeling the image as a spatial-temporal sequence and its shape was used as input for the RNN:

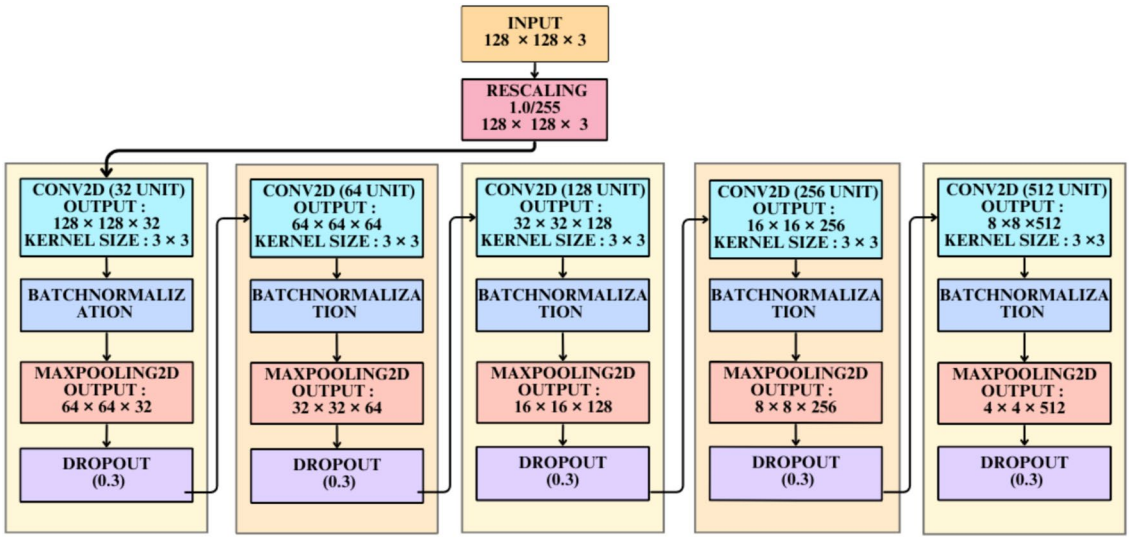


Fig. 6. CNN branch of C-RNet model.

$$P \in \mathbb{R}^{256\times 192}$$

As a result, each patch was processed individually, which helped the model to learn features faster, more easily and efficiently. Following that, a Bidirectional LSTM processes the patches in both forward and backward directions:

Forward pass

$$\overrightarrow{h_t} = \mathrm{LSTM}(P_t, \overrightarrow{h_{t-1}})$$

Backward pass

$$\overleftarrow{h_t} = \mathrm{LSTM}(P_t, \overleftarrow{h_{t+1}})$$

The final representation is:

$$v_{\mathrm{RNN}} = \left[\overrightarrow{h_T} \,\|\, \overleftarrow{h_T}\right]$$

The output of the bidirectional LSTM layer was then fed into a dense layer with 128 units and the ReLU activation function. This layer converted the LSTM output into a feature vector prepared for further processing. L2 regularization was utilized to avoid overfitting. Finally, a dropout layer was applied, randomly deactivating 30% of the neurons to reduce overfitting.

Figure 7 provides an overview of the RNN branch of our model.

CNN and RNN concatenation

The Concatenation layer, the final component of C-RNet, demonstrates its primary architectural innovation and is essential to improving the hybrid model's performance.

The CNN branch for processing image data and the RNN branch for handling sequential data. These branches were designed to extract meaningful features from their respective inputs before merging into a unified representation for the final classification task.

Let $v_{\mathrm{CNN}} \in \mathbb{R}^{256}$ denote the global spatial descriptor and $v'_{\mathrm{RNN}} \in \mathbb{R}^{128}$ denote the sequential structural descriptor. In the architecture, CNN learns local texture features (like edges, opacities, nodule shapes, gradients etc.) and Bi-LSTM learns global structural dependencies (like bilateral symmetry, multi-lobe patterns, sequential

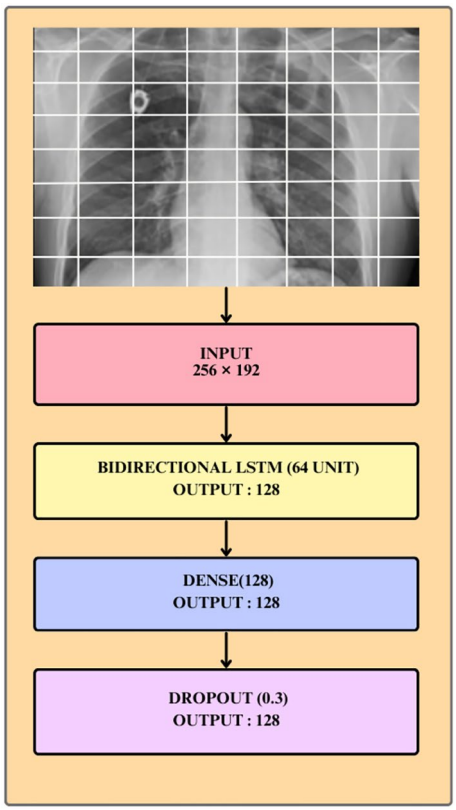


Fig. 7. RNN branch of C-RNet model.

transitions etc.)[35]. Concatenating the embeddings forces the model to jointly reason about fine-grained details (CNN) and long-range anatomical context (RNN). To unify spatial and sequential information, the two learned representations are concatenated to form the fused embedding:

$$v_{\text{fusion}} = v_{\text{CNN}} \parallel v'_{\text{RNN}}$$

Simultaneously, the concatenation method seamlessly combines spatial data from images and temporal data from sequences and thus is ideal for multimodal data-intensive applications. Moreover, the model is designed to be robust and durable by including regularization techniques such as L2 regularization and dropout layers. This improves its performance with previously unknown data.

Explicitly, the fused representation is:

$$v_{\text{fusion}} = \left[v_{\text{CNN}}^{(1)}, \ldots, v_{\text{CNN}}^{(256)}, v_{\text{RNN}}^{(1)}, \ldots, v_{\text{RNN}}^{(128)} \right]$$

This fused vector is passed through a Dense layer of 128 ReLU units, integrating spatial and sequential features. This layer encourages interaction between CNN and LSTM features, enabling the network to learn hybrid spatio-temporal patterns.

Finally, a softmax classifier outputs probabilities for the four disease categories:

$$\hat{\mathrm{y}} = \mathrm{softmax}\left(\mathrm{W}_o \mathrm{z} + \mathrm{b}_o\right)$$

Moreover, this fusion method preserves the complete representational power of each branch intact, avoiding the rush to merge them into a single feature space prematurely. In addition to being theoretically viable, this fusion technique is computationally effective and essential to attaining high test accuracy. This is especially crucial when we have to interpret contextual and spatial cues simultaneously, such as when differentiating between viral pneumonia, lung opacity, and normal. Fig. 8 provides a visual of the integration between CNN and Bi-LSTM concat of our C-RNet model.

Experimental evaluation

Experimental setup

The CNN-RNN model suggested was developed and validated in an environment that was equipped with Windows 11 Pro[36] version: 24H2, an Intel Core i7 14th Generation processor, 64 GB RAM, 1 TB storage, and an NVIDIA GeForce RTX 4080 SUPER GPU. The GPU supported CUDA version 12.9[37] with driver version 576.57[38]. We conducted all experiments using Python 3.12.7[39] in a Jupyter Notebook 7.2.2[40] environment. The deep learning models were implemented with TensorFlow 2.18.0[41] and Keras 3.8.0[42]. The python libraries employed included NumPy 1.26.4[43], Panda 2.2.2[44], Matplotlib 3.9.2[45] and OpenCV 4.11.0[46]. We managed packages using Pip 24.2[47] and Conda 24.11.3[48]. For image processing tasks, we used Scikit-image 0.24.0[49].

Evaluation metrics

To ensure a robust assessment beyond a single measure, we have utilized multiple metrics including accuracy, precision, recall and F1-score. These metrics comprehensively evaluate our model's performance. The various evaluation metrics utilized in this research can be represented through the following equations:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

Here, TP stands for correctly predicted positive cases or true positives. TN is correctly predicted for negative cases. FP and FN are false positives and false negatives. These are the cases that are incorrectly predicted. Although accuracy is a widely used metric, it does not always reflect the model's true performance, especially in imbalanced datasets. By incorporating precision, recall, and F1-score, we ensure a more reliable evaluation that considers both false positives and false negatives. These metrics help in optimizing the model for real-world applications where misclassification costs vary depending on the context.

Performance analysis

The analysis result represented in Table 1 provides significant details regarding how well the C-RNet model classified lung diseases. The model exhibited exceptional precision, recall and F1-scores in four categories: COVID, Lung Opacity, Normal and Viral Pneumonia. With 98% precision and 95.4% recall, the COVID class performs extraordinarily well, according to the classification results, generating an F1-score of 96.7%. This illustrates how accurately the model detects COVID instances with little false positives. With a recall rate of 89.7% and an accuracy of 91.6%, the lung opacity F1-score is still remarkably high at 90.7%. However, the minor

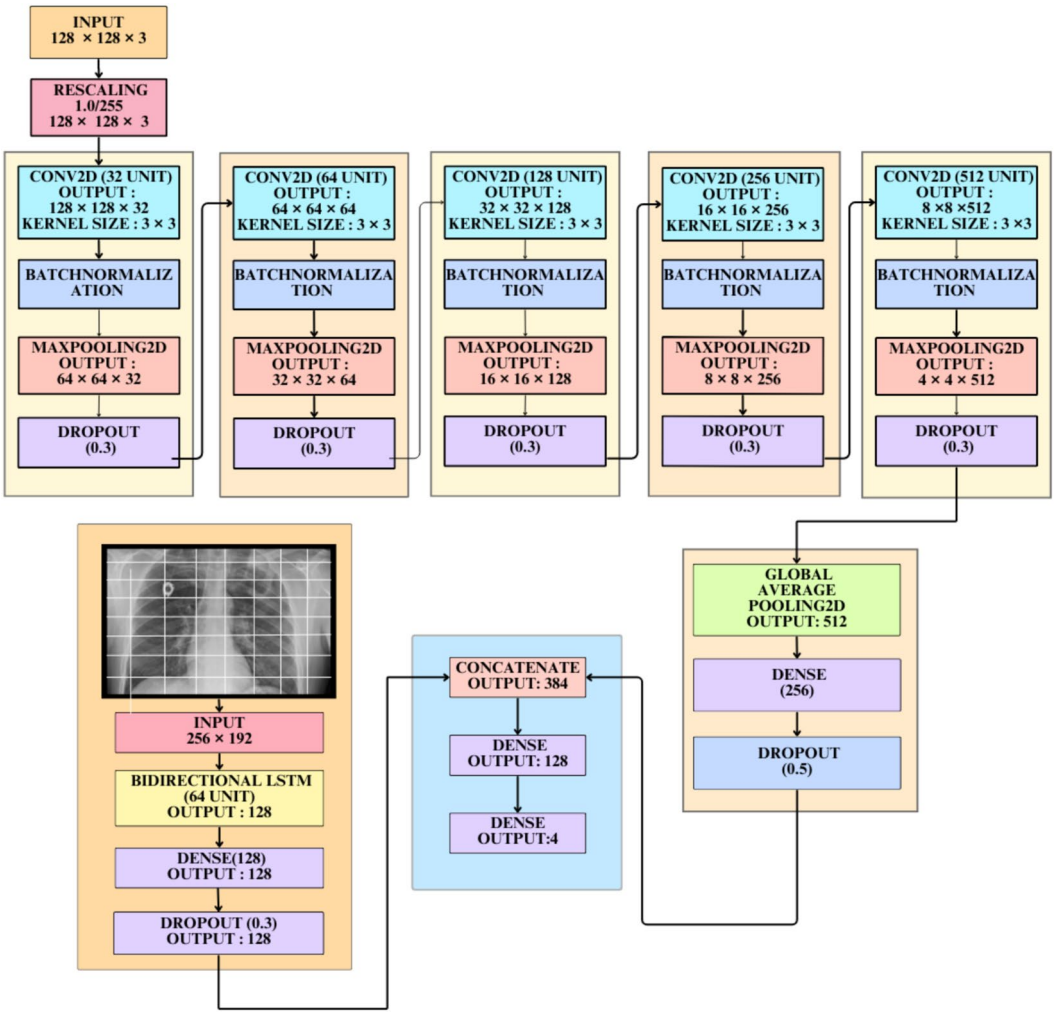


Fig. 8. C-RNet model.

Class	Precision	Recall	F1-Score	Support
COVID	0.980	0.954	0.967	1080
Lung Opacity	0.897	0.916	0.907	1835
Normal	0.942	0.941	0.942	3049
Viral Pneumonia	0.974	0.966	0.970	386
Accuracy			0.937	6350
Macro Avg	0.948	0.944	0.946	6350
Weighted Avg	0.938	0.937	0.937	6350

Table 1. Performance metrics value of the C-RNet model.

drop in accuracy shows that the model might incorrectly classify certain situations, defining an area that should be upgraded.

With 94.2% precision and 94.1% recall, the Normal category performs well overall, generating a 94.1% F1-score. In medical diagnostics, this is especially important because correctly identifying normal situations lowers the chance of unwanted medications. The model performs exceptionally well for viral pneumonia, achieving 97.4% precision and 96.6% recall with a 97% F1-score. This shows how well the model can identify cases of viral pneumonia while minimizing false negatives. The C-RNet model's average accuracy of 93.7% demonstrates its dependability and potency as a medical imaging diagnostic tool. This great performance is further supported by its high weighted average and macro ratings. In addition to the metrics listed above, the model had a Macro AUC of 0.992 and a Weighted AUC of 0.989, confirming its strong discriminative abilities across all four classes.

Furthermore, our model surpassed larger designs with a total of 1.7 million parameters and a compact size of 7.25MB. In real-time scenarios, memory as well as speed are essential. These high-performance demonstrations emphasize the strength of our spatial-sequential feature fusion approach.

Figure 9 presents the confusion matrix of our proposed model for lung cancer detection, which consists of four labels: "Viral Pneumonia", "Normal", "Lung_Opacity" and "COVID". From the confusion matrix, it is evident that the model classifies the majority of COVID, Normal and Viral Pneumonia cases correctly, with only a few exceptions. COVID and Viral Pneumonia indicate high accuracy and are rarely mistaken with other classes. With only a few small overlaps into Lung Opacity, the model displays an excellent job of differentiating COVID-19. Due to similar patterns, Lung Opacity overlaps with Normal. This is expected given that early or mild opacity resembles normal lung texture and dense opacities can overlap radiologically with COVID patterns. Despite C-RNet's excellent overall performance, the most misclassification happens between Lung Opacity and Normal. Lung opacities can appear as patchy patterns, slight haziness. Hence, the model considers it more difficult to learn a single compact representation.

Furthermore, occasional false positives in pneumonia detection may occur, indicating cases in which the model need further improvement. By recognizing these constraints, we gain a realistic assessment of the model's clinical applicability and highlight potential areas for future improvement. Overall, most correct predictions lie along the diagonal, implying that the model performs well in identifying lung diseases from X-rays. It is promising for practical medical use in lung disease identification, utilizing X-ray pictures, which is highlighted by its success across all classes, particularly in accurately differentiating between various lung diseases.

Impact of dataset imbalance

Despite the imbalancing in the dataset, the proposed C-RNet model demonstrates remarkable generalization across all four categories due to several architecture factors. Firstly, the fusion mechanism captures both global image level and fine-grained patch level features. This reduces an excessive dependence on patterns from the majority class. Additionally, the patch-based sequence method increases data variability by dividing each image into 256 patches. This compensates for limited minority class samples, provides a large number of learning signals and acts as a structural form of augmentation. Moreover, regularization techniques like L2 regularization, dropout and batch normalization support the model's capability to learn distinguishing features as demonstrated by the high recall of the minority classes (COVID-19 = 0.96, Viral Pneumonia = 0.97).

To further validate, we performed an experiment using class-weighted training. The weights were:COVID (0): 1.46, Lung Opacity (1): 0.88, Normal (2): 0.52, Viral Pneumonia (3): 3.86. As a result, sensitivity for minority classes improved but total accuracy fell to 82.81%. Notably, Normal class significantly decreased from 0.94 to 0.67. This outcome reinforces that the model is already managing imbalance effectively. Artificially balancing the dataset ultimately affects the model's real-world performance.

Training convergence comparison

To evaluate how well our model learns, generalized and remained stable, we compared its training and validation accuracy over 100 epochs with four other models- Xception[50], VGG19[51], ViT[52], Inception LSTM[52]. The accuracy of all models were visually represented in the Figures 10a , 10b , 10c , 10d and 10e .

In the Fig. 10a , the C-RNet model demonstrated consistency in learning from training data. The training accuracy rose steadily, reaching approximately 96%, indicating the model had effectively mastered the training set. The validation accuracy displayed some inconsistent patterns in the learning phase, initially indicating instability in generalization. However, as time passed, the validation accuracy began to stabilize. It began to align more closely with the training accuracy line after around 40 epochs. From that point onward, the validation

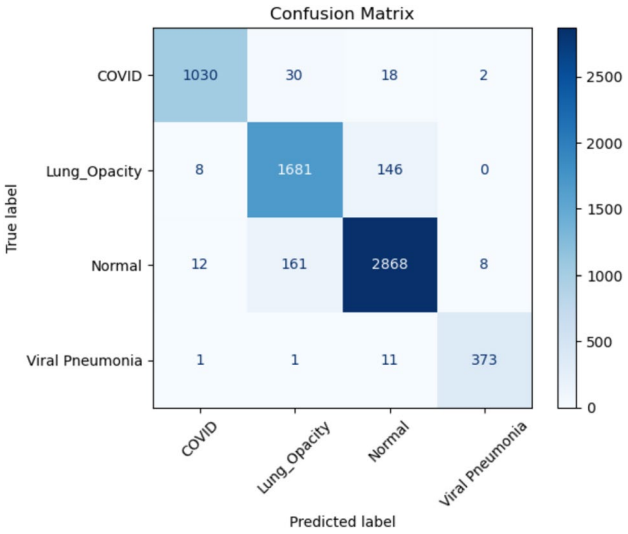
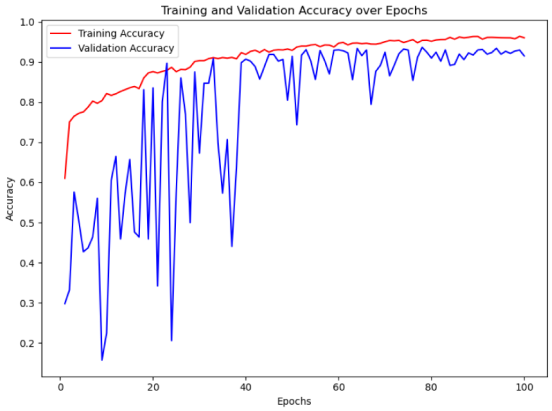


Fig. 9. Confusion matrix of C-RNet model.



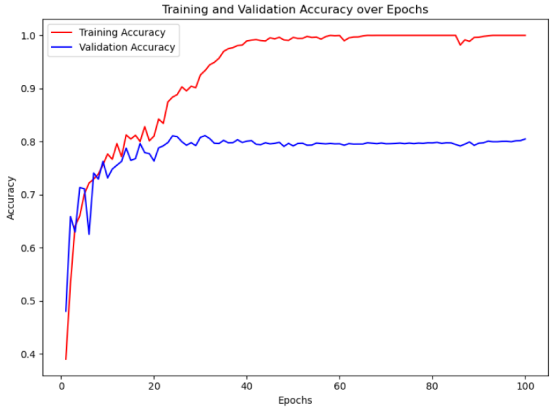
(a) C-RNet



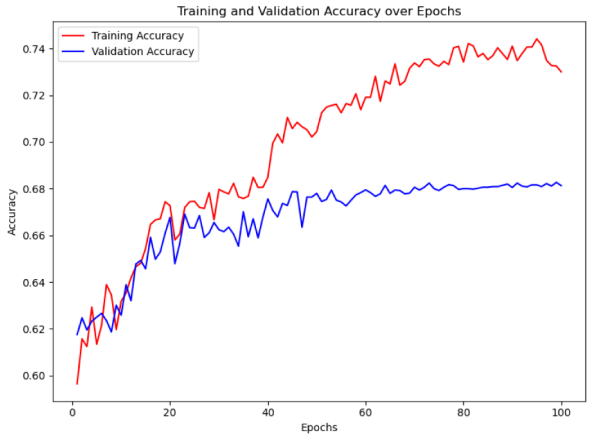
(b) Xception



(c) VGG19



(d) ViT



(e) Inception LSTM

Fig. 10. Accuracy vs epoch graph.

accuracy continuously stays above 85% and hovers around 90% with just slight variations from epoch 40 onward. The model started to generalize well to unknown input after progressively overcoming early learning noise. This convergence indicated that the model was learning meaningful features and also generalizing well to unseen data, while continuing to improve training performance. These accuracy values also implied low overfitting and strong convergence.

In contrast, the other models achieved high values in training accuracy, but the validation accuracy plateaued much earlier. The noticeable gap between training and validation accuracy in all models suggested the models were struggling to generalize and overfitting during evaluation. Among them, Fig. 10b , Xception performed best with a training accuracy of 94.50%. The VGG19 model (Fig. 10c) showed a sharp rise in training accuracy, but the gap between the accuracies indicated weak generalization. The ViT model, shown in Fig. 10d , appeared

sensitive to data as it reached around 80% validation accuracy relatively quickly and remained flat. In contrast, the Inception LSTM model (Fig. 10e) underperformed overall, with consistently lower training and validation accuracy compared to the other models.

Overall, our study provided a complete convergence comparison of CNN-based models for lung disease classification from X-rays, highlighting the proposed C-RNet's improved accuracy and efficiency, as well as its potential for improving automated medical diagnostic tools. Our C-RNet model achieved higher validation accuracy than the other models, indicating broad generalization and reliability for real-world clinical applications where reliable performance on previously unknown data is critical. Its better performance in validation accuracy than other models makes the model more suited for deployment.

CNN branch with sequential attention block(SE)

To further enhance discriminative feature learning, we conducted an experiment with a Squeeze-and-Excitation (SE) attention mechanism. We selectively integrated SE blocks into deeper convolutional layers (128, 256, and 512 filters). This design decision prevents overfitting and unnecessary parameter development in previous layers while enhancing channel-wise feature discrimination. The SE blocks provide channel-wise feature recalibration by accentuating informative feature maps and diminishing less pertinent ones, which is very beneficial in medical imaging activities that involve delicate disease patterns[53].

In contrast to intensive attention strategies, SE attention ensures that the total model size stays below 50 MB by adding very few extra parameters. The goal of the study is to create an effective and deployable architecture without sacrificing interpretability or performance, and this design supports that goal. Figure 11 provides an visualization of the integration of SE attention block in the CNN branch of our proposed C-RNet model.

Results comparison

This section provides a fair and holistic assesment of the C-RNet architecture using multiple evaluation metrics-accuracy, precision, recall. F1-score, model size and training time. The comparison is conducted against several widely used deep learning models, including Xception, Vision Transformer (ViT), VGG19, Inception-LSTM, and C-RNet+SE, for multi-class lung disease classification from chest X-ray images. Among these, our proposed C-RNet model performed noticeably better across every major evaluation metric. Table 2 demonstrates the accuracy value of all the models.

To highlight the advantages of our proposed model, we compare the behaviour of the model with transformer based models and traditional single-branch CNNs. With the test accuracy of 93.73% and a training accuracy of 97.90%, the minimal generalization gap showed that C-RNet model learns features effectively without overfitting. The highest F1-score of 94.6% indicates that it also achieved a superior balance between precision and recall, which is essential for guaranteeing accurate classification in actual clinical settings. Overall, C-RNet presented a better trade-off between generalizability, accuracy, and model size, making it an optimal solution for high-performance and lightweight

In contrast, traditional CNN architectures like VGG19, Xception and Inception only extract only local spatial features. Thus, long-range contextual patterns like zone-to-zone progression or multi-focal infiltrates remain uncaptured by this model, which are crucial for interpreting chest X-rays. Although Xception and VGG19

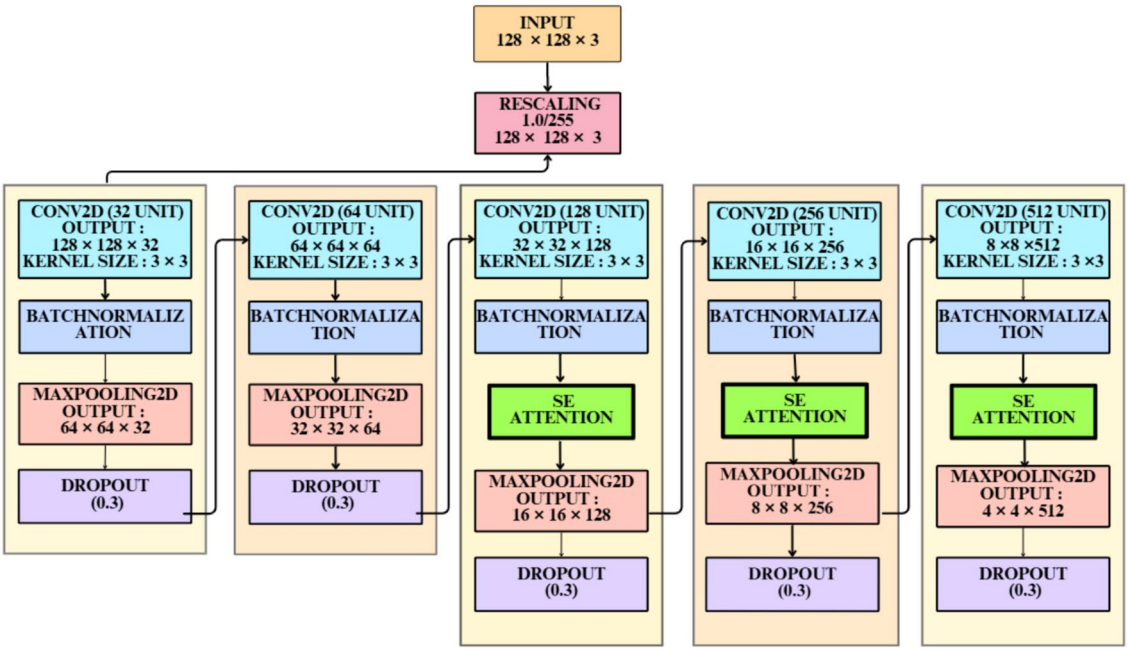


Fig. 11. CNN branch with SE attention block.

Model	Train Acc. (%)	Test Acc. (%)	Precision (%)	Recall (%)	F1-score (%)	Model size	Train time(s/epoch)
C-RNet	97.90	93.73	94.8	94.4	94.6	7.25 MB	216.81
Xception	94.85	92.63	93.03	92.65	92.82	81.59 MB	539.5
ViT	92.48	82.57	81.71	81.52	81.57	26.09 MB	707.73
VGG19	95.12	88.28	91.59	92.12	91.83	82.89 MB	625.39
Inception LSTM	74.48	68.27	68	66	67	91.64 MB	112.23
C-RNet+SE	92.16	89.27	89	94	91	7.42 MB	112.23

Table 2. Comparison of models performance.

Model	Classes	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Nair et al.[22]	6	89.31	95	99	94
Saky et al.[54]	5	98.15	90.65	89.27	89.90
Patil et al.[55]	1	76.02	94.81	84.34	90.1
Proposed C-RNet Model	4	93.51	94	93.70	94

Table 3. Comparison of results obtained in this study with other methods in the literature.

displayed exceptional test performance, their large model sizes limited their practicality for deployment in resource-constrained clinical environments.

The Vision Transformer(ViT) model achieved a test accuracy of only 82.57% and an F1-score of 81.57%. These value exhibit weaker generalization performance despite having the capabilities of modeling global context. This performance drop is due to ViT's high data and computational requirements, which make it less suitable for modestly sized medical datasets. Similarly, the Inception-LSTM model performs weakly on all evaluation parameters, indicating that naively combining CNN and recurrent layers without effective feature fusion is insufficient robust lung disease classification.

Furthermore, from an efficiency perspective, C-RNet demonstrates a strong advantage. The model maintained a compact model size of only 7.25 MB, which was significantly less than the other models., including Xception (81.59 MB) and VGG19 (82.89 MB). This efficiency made C-RNet particularly suitable for deployment in resource-constrained environments. Additionally, C-RNet offers a lightweight alternative for capturing long-range relations, achieving 99.19% macro accuracy with only 1.9M parameters, whereas ViT displays weaker generalization. While Xception and VGG19 demonstrated outstanding test performance, their much higher model sizes gave rise to deployment issues. ViT scored moderately despite its tiny size, but Inception LSTM underperformed across all parameters, particularly generalization(F1-score of 68.27%). This demonstrated that by hybridization technique, it is viable to achieve stronger global reasoning without the computational and data requirements of transformer architectures.

In terms of training time efficiency, C-RNet achieves an excellent balance with a training time of only 216.81 seconds per epoch. It trains significantly faster than heavier models like ViT, which takes 707.73 seconds per epoch and VGG19, which takes 625.39 seconds per epoch. Although Xception may train faster, but it is less accurate than C-RNet and has a model size that is more than 10 times greater (81.59 MB). This time study demonstrates our architecture's practicality and computational efficiency. .

To analyze the influence of channel-wise attention, we assessed a C-RNet variation that included SE blocks in deeper convolutional layers (C-RNet+SE). As shown in Table 2, C-RNet+SE obtained a test accuracy of 89.27% and an F1-score of 91%, which are lower than the baseline C-RNet (93.73% accuracy and 94.6% F1-score), despite having a slightly larger model size (7.42 MB vs. 7.25 MB). Although SE attention strengthened recall (94%), it impacted precision (89%), causing more false positives, especially in visually overlapping classes like Lung Opacity and Normal. While C-RNet+SE improves training efficiency (112.23 s/epoch), it compromises generalization performance. Overall, our findings suggest that explicit spatial-sequential modeling via the Bi-LSTM branch outperforms channel-wise attention solely for this task, validating the choice of the baseline C-RNet as a suitable trade-off between accuracy, robustness, and deployment efficiency.

Moreover, we have compared the performance of our proposed model's findings with several recent studies. In Table 3 displays this comparison across key evaluation metrics. This comparison demonstrates the relevance and effectiveness of the proposed architecture in relation to current state-of-the-art approaches.

Relationship between model size and test accuracy

The proposed C-RNet model occupies 7.25MB and contains 1,901,764 trainable parameters when using 32-bit floating-point precision. The model obtained a strong accuracy of 93.73% despite its compact architecture. This demonstrates an effective balance between model complexity and predictive performance. This efficiency arises from the model's integration of complementing elements.

In the architecture, the CNN branch is designed to extract spatial hierarchies through five convolutional blocks. However, architectural choices like Global Average Pooling, L2 regularization and Dropout help prevent excessive parameter inflation. On the other hand, the RNN branch operates with image patches, enabling a Bidirectional LSTM to effectively capture long-range dependencies without expanding the parameter count.

Each LSTM unit contributes in keeping the recurrent module computationally light by maintaining a modest hidden size. As a result, this hybrid approach enhances feature learning without considerably increasing model size.

From a deployment perspective, the compact size of C-RNet ensures that it's a great fit for low-end GPUs, older CPUs, microservers and budget devices where memory and processing capabilities are limited. This feature is particularly beneficial in real-world scenarios like medical settings, fieldwork or mobile applications where high-performance hardware may be unavailable. This relationship showcases the model's strong performance relative to its size, which is essential for running on lower-end or resource-limited hardware.

Grad-CAM visualization analysis

To improve the evaluation of our exclusive CNN-RNN model, particularly for lung disease classification from CXR images, we have used Grad-CAM in our study. Grad-CAM provides visual explanations by identifying and analyzing the specific regions of chest X-ray that have a significant effect on the model's predictions. This method displays how particular picture attributes influence the CNN-RNN model's decision-making process.

In our approach, we generated heatmaps that illustrated the most important regions in the X-ray pictures by calculating the gradients of the projected class in relation to the feature maps of the corresponding layer. Heatmaps also explains where the model "focuses" when identifying abnormalities. The heatmaps highlighted key regions that influence predictions, with red and yellow suggesting high activation (possible abnormalities) and blue and green indicating low activation. We were then able to visibly connect the model's concentration with real lung features by superimposing this heat map across the initial X-ray image.

The interpretability and feature localization of C-RNet differ significantly from baseline designs such as Xception, VGG19, ViT and Inception-LSTM. C-RNet reliably produced targeted, anatomically relevant activations inside the lung fields, especially around the areas affected by infection. In COVID-19 and Viral Pneumonia cases, C-RNet accurately focused on peripheral ground-glass opacities and patchy infiltrates. In contrast, VGG19 and Xception exhibited fragmented or dispersed activation patterns frequently extending into surrounding soft tissues. This fragmented activation suggested that these models might be detecting non-diagnostic cues or noise instead of clinically important lung characteristics, which could produce false results in practical applications.

Furthermore, ViT-generated heatmaps were slightly diffused, implying that they lack the spatial precision required to identify localized radiographic abnormalities. Similarly, Inception-LSTM and Xception generated heatmaps that were incorrectly diverted from the actual lung pathology toward the shoulders or diaphragm, indicating evidence of feature leakage. These problems drew attention to localization difficulties and raised concerns about the model's reliability in actual clinical settings.

In contrast, C-RNet showcased stability and well-localized attention in every class. C-RNet produces sharper, more focused heatmaps over abnormal spots. C-RNet constantly activated the appropriate areas, demonstrating its supremacy in feature extraction and kept robust attention across diverse X-rays in comparison to other models, demonstrating reliability. While C-RNet created distinct, disease-focused heatmaps, other models displayed dispersed, irrelevant activations.

These visual comparisons showed that C-RNet offered the most accurate and reliable Grad-CAM visualizations, producing sharper and anatomically aligned heatmaps with stronger clinical alignment, fewer false activations and improved abnormality localization. This method was essential for confirming the model's precision and dependability. In medical imaging, where comprehending the rationale behind a model's judgments is as crucial as attaining high accuracy, Grad-CAM increases confidence in the CNN-RNN model by visually emphasizing significant areas and making its predictions clearer. The Grad-CAM visualization of our proposed model could be seen in the Fig. 12a, b, c, d and e.

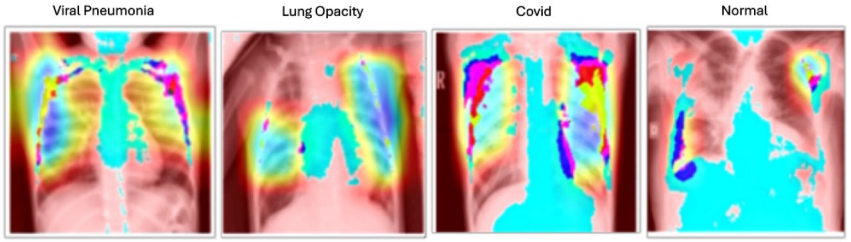
Discussion

Key findings

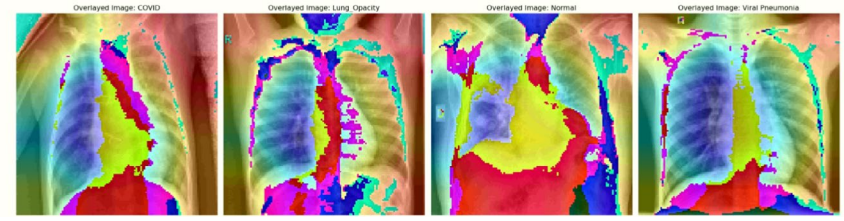
Our proposed model classifies lung conditions, including COVID-19, Lung Opacities, Normal, and Viral Pneumonia by integrating CNN and BiLSTM networks. The CNN extracts spatial features from CXR images, and BiLSTM captures sequential pattern recognition. Traditional studies have generally used CNNs for feature extraction or fed CNN-derived features as RNN inputs[56]. In contrast, our proposed model improves performance by processing entire images in the CNN branch while supplying 256 image patches as time-series data to the BiLSTM. The primary outcomes of our study indicate the efficiency of our C-RNet model in classifying lung diseases. This dual-stream architecture considerably increases the model's robustness to varying image resolutions and noise, making it more versatile across diverse datasets. The CNN module captures local spatial features, ensuring that complicated patterns in radiographic images are adequately represented. Furthermore, the BiLSTM branch improves performance by evaluating the dependencies between extracted features, allowing the model to grasp temporal interactions. This combination allows the RNN to identify variations in pixel flow, which is essential for disease detection. Overall, our findings demonstrate that C-RNet is a viable tool for lung disease diagnosis and directly address the research question regarding effective integration of spatial and sequential learning.

Advantages of C-RNet over existing methods

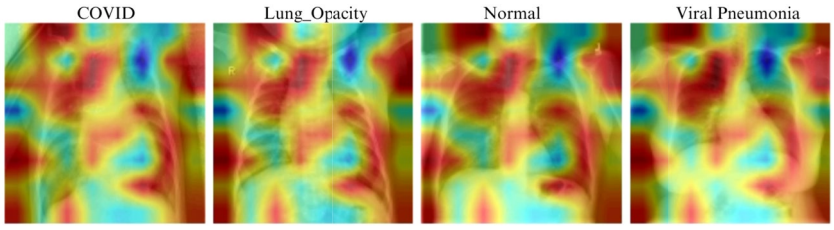
Although there are several deep learning models for medical image classification. Our model differs from others due to its importance in simultaneously extracting sequential and spatial features. Most traditional CNN-based models rely on spatial feature extraction without considering contextual relationships[57]. Typically, CNNs extract features from images and pass them to RNNs as input. In this method, sequence size matching is required, and sometimes spatial relationships can be lost due to the sequence size sorting. In contrast, our model incorporates



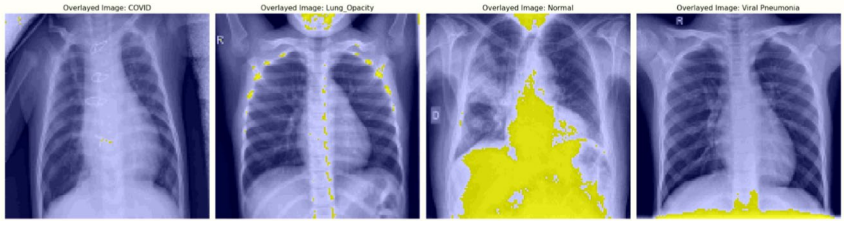
(a) CRNet Model



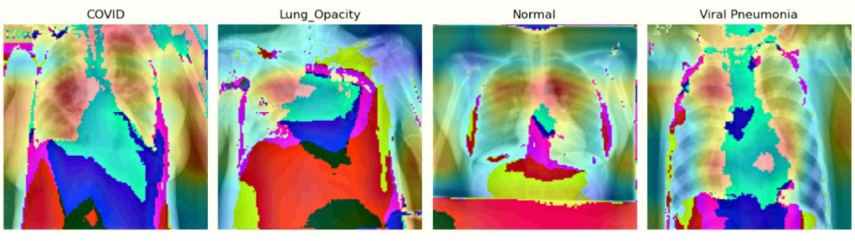
(b) Xception



(c) ViT



(d) Inception LSTM



(e) VGG19

Fig. 12. Grad-CAM visualization of different models.

an advanced patch-based augmentation technique that improves generalization across different imaging conditions. Hong et al.[58] also proposed a multi-class classification method of lung disease using a CNN model. But, their model only collected CXR pictures that are only in the PA (Posteroanterior) and AP (Anteroposterior) positions, with no indication of the disease's exact location. In contrast, our C-RNet model solves this restriction by using Grad-CAM visualization, which highlights specific parts of the X-ray where abnormalities are found. This not only enhances interpretability but also aids in clinical decision-making by providing visual evidence of impacted areas. As a result, our model improves both classification accuracy and explainability, making it better suited for real-world medical applications. The study by Sulaiman et al.[59] does not address picture quality variability or the resemblance between normal and abnormal lung areas, and the ideal choice of k for a CNN model depends on variables such as dataset size. Our approach, on the other hand, improves its robustness

in a variety of scenarios by handling image quality variations and performing well across several datasets[60]. Moreover, another study by Khan et al.[61] performs well, but it requires a lot of processing power, which can make it less useful in actual clinical settings. Our C-RNet model, on the other hand, is made with computational efficiency in mind, guaranteeing quicker processing without sacrificing accuracy. This makes our method better suited for implementation in healthcare environments with limited resources, where efficiency and performance must be balanced. However, while our C-RNet Model performed well in some aspects, there is still some scope for further improvement in future work. Future work should explore incorporating additional pre-processing techniques such as histogram or various filters and segmentation-based feature extraction to improve model input quality. Additionally, integrating multi-modal data sources, including CT scans, clinical metadata, and laboratory results, can significantly improve classification performance.

Conclusion

In underdeveloped nations like Bangladesh, where access to skilled radiologists and cutting-edge diagnostic facilities is restricted, respiratory illnesses continue to be a significant global public health risk. Therefore, enhancing treatment results and lowering deaths depend on timely and precise diagnosis of lung disorders employing affordable imaging methods, notably chest X-ray (CXR). In order to improve computational lung disease categorization from CXR pictures, this research presented a combined deep learning model called Convolutional Recurrent Network (C-RNet).

Key contribution

We created a customized deep-learning architecture that successfully combines CNN for global feature collection and RNN for sequential reliance training in order to improve the precision and effectiveness of lung disease diagnosis. Our work employed a publicly accessible dataset to categorize CXR pictures into four different groups.

The constraints of conventional single-scale CNN-only techniques are overcome by this dual-stream design, which enables the model to concurrently absorb extended spatial correlations and local structural features. The suggested approach improves the precision of classification by effectively capturing the continuity and dependency characteristics of subsequent levels. Patch-based analysis enhances generalization significantly while lowering the requirement for explicit data augmentation.

Comprehensive tests on the publicly accessible COVID-19 Radiography Database show that C-RNet outperforms a number of cutting-edge models, such as Xception, VGG19, Vision Transformer (ViT), and Inception-LSTM. With a 93.73% rate of success on the entire dataset, our model displays the potential to enhance diagnostic performance. The primary finding of this research demonstrates that the combined strategy functions better than conventional single-scale techniques. Overall precision, recall, and F1-score of 94.6% further confirm the effectiveness of the proposed approach.

Our suggested C-RNet model's compact size of just 7.25 MB is another significant advantage; it is noticeably smaller than other pretrained models that we trained and assessed on the same dataset. Additionally, the model's cheap computational cost and lightweight design make it ideal for use in clinical settings with limited resources. Its lightweight design makes it perfect for real-world clinical use in Bangladesh's rural areas, where processing power and clinical resources are limited.

Furthermore, by offering graphical descriptions of affected areas in CXR images, the insertion of explainable AI (XAI) utilizing Grad-CAM significantly improves the clinical dependability of the model. When taken as a whole, these contributions demonstrate that C-RNet is a lightweight, efficient, and comprehensible method for diagnosing multiple classes of lung diseases.

Limitations and future work

Despite its notable performance, C-RNet has distinct limitations. The primary weakness is the false positives cases in Lung Opacity. Normal images are occasionally wrongly classified as mild opacity. This usually happens in borderline or low-contrast conditions where radiographic features are very delicate. Similarly, small samples of Viral Pneumonia are categorized as Normal due to mild inflammatory patterns. These false cases reflect the actual diagnostics challenges encountered in chest radiography and the struggle of high performing models in classifying subtle image variations. Another drawback is the diversity in CXR image quality across the dataset, which includes differences in noise, sharpness, contrast and exposure. While C-RNet has showcased strong generalization, low-quality pictures might mask minor radiographic features, resulting in borderline misclassifications. Under low light, mild opacities may resemble normal lung tissue, and early-stage viral pneumonia patterns may look too faint for reliable diagnosis. Such image quality fluctuations present issues for both automated models and human radiologists, highlighting a fundamental drawback of clinical datasets.

To address these limitation, future research will concentrate on strengthening the suggested model's robustness to varying image quality. Noise reduction preprocessing and quality-aware data augmentation techniques are among the planned initiatives, which replicate under-exposed, low-contrast and artifact-affected photos. Moreover, We plan to use different optimizers and other data augmentation techniques in an attempt to further improve the classification accuracy of the proposed CNN architecture with data augmentation. We also plan to use early stopping and batch normalization instead of dropout layer to see their effect in avoiding overfitting.

Reproducibility and replicability

The complete implementation of the proposed CNN-RNN architecture, including preprocessing, model building, training and evaluation has been made available via the following public GitHub repository with necessary dependencies and documented scripts, https://github.com/ZahinZee/Enhancing_Lung_Diseases_Recognition_Through_CNN-RNN_Methodologies. The framework is designed to be executed in any standard

GPU-enabled environment using tensorflow and Keras, enabling independent validation and extension of the suggested method.

Data availability

The datasets analysed during the current study are available in the Kaggle repository, https://www.kaggle.com/datasets/tawsifurrahman/covid19-radiography-database

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Author contributions

I.A.Z.: Literature Search, Theoretical Contextualization, Implementation, Validation, Writing - original draft, Reviewing and Editing (full manuscript correction). M.F.A.: Conceptualization, Data curation, Implementation and Software. R.A.O.: Formal Analysis, Results Evaluation, Writing - original draft, Designing Figs. 1, 4 and 11. A.H.: Methodology, Designing Figs. 3, 6, 7 and 8. M.F.A.: Literature Search. M.A.O.Z. and J.N.: Supervision, Expert Guidance and Reviewing. The finished manuscript was examined and approved by all authors.

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Declarations

Competing interests

The authors declare no competing interests.

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